INTRODUCTION

We have already learnt the chemical equilibrium in XI standard. In this unit, we discuss the ionic equilibria, specifically acid – base equilibria. Some of the important processes in our body involve aqueous equilibria. For example, the carbonic acid – bicarbonate buffer in the blood.

$$H_3O^+(aq)+HCO_3^-(aq) \rightleftharpoons H_2CO_3(aq)+H_2O(l)$$

We have come across many chemical compounds in our daily life among them acids and bases are the most common. For example, milk contains lactic acid, vinegar contains acetic acid, tea contains tannic acid and antacid tablet contains aluminium hydroxide / magnesium hydroxide. Acids and bases have many important industrial applications. For example, sulphuric acid is used in fertilizer industry and sodium hydroxide in soap industry etc… Hence, it is important to understand the properties of acids and bases.

In this unit we shall learn the definitions of acids and bases and study, their ionisation in aqueous solution. We learn the pH scale and also apply the principles of chemical equilibrium to determine the concentration of the species furnished in aqueous solution by acids and bases.

8.1 Acids and bases

The term 'acid' is derived from the latin word *'acidus'* meaning sour. We have already learnt in earlier classes that acid tastes sour, turns blue litmus to red and reacts with metals such as zinc and produces hydrogen gas. Similarly base tastes bitter and turns red litmus to blue.

These classical concepts are not adequate to explain the complete behaviour of acids and bases. So, the scientists developed the acid – base concept based on their behaviour.

Let us, learn the concept developed by scientists Arrhenius, Bronsted and Lowry and Lewis to describe the properties of acids and bases.

8.1.1 Arrhenius Concept

One of the earliest theories about acids and bases was proposed by swedish chemist Svante Arrhenius. According to him, an acid is a substance that dissociates to give hydrogen ions in water. For example, HCl, H_2SO_4 etc., are acids. Their dissociation in aqueous solution is expressed as

$$HCl(g) \xrightleftharpoons{H_2O} H^+(aq)+Cl^-(aq)$$

The H^+ ion in aqueous solution is highly hydrated and usually represented as H_3O^+, the simplest hydrate of proton $\left[H(H_2O)\right]^+$. We use both $H^+$and H_3O^+ to mean the same.

Similarly a base is a substance that dissociates to give hydroxyl ions in water. For example, substances like NaOH, $Ca(OH)_2$ etc., are bases.

$$Ca(OH)_2 \xrightleftharpoons{H_2O} Ca^{2+}(aq)+2OH^-(aq)$$

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